Remote Sensing Scene Classification Based on Attention-Enabled Progressively Searching

Junge Shen, Bin Cao, Chi Zhang, Ruxin Wang, *Member, IEEE*, and Qi Wang, *Senior Member, IEEE*

***Abstract*— Remote sensing image scene classification plays a significant role in remote sensing image analysis. Aiming at the problems of large transformation and scale variation of background and key objects in remote sensing images, we propose a neural architecture search (NAS) method based on attention search space. The network adaptively searches convolution, pooling, and attention operations in the appropriate layers. To ensure the stability of the searching process, a multistage network progressive fusion search method is proposed, which discards useless operations in stages, reduces the burden of search algorithm, and improves the search efficiency. Finally, paying attention to the association information between objects and scenes, a bottom-up multiscale fusion network connection strategy is proposed to fully reuse the semantics of multiscale feature maps in each stage. The experimental results show that the proposed method performs better than the manual method and the current neural network architecture search method.**

***Index Terms*— Convolutional neural network (CNN), feature fusion, neural architecture search (NAS), remote sensing scene classification.**

I. Introduction

THE rapid development of remote sensing devices and techniques enables us to acquire high-quality remote sensing data in an increasing speed [1]. Concurrently, an important task arises, which regards automatically exploiting the meaningful and useful information from these data. Remote sensing scene classification is a typical research topic that focuses on understanding the content of the scene images and correspondingly assigns a human-preferred label for each image. Advanced techniques could facilitate the processing of a large amount of remote sensing images, which have already been attended in practices, such as agriculture [2], military [3], environment [4], and geography [5].

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The complex content in remote sensing images poses a great challenge to the high-level vision tasks including scene classification. Conventional hand-crafted feature extraction methods cannot fulfill the strict performance requirement on the increasing remote sensing data, which is mainly because of the insufficient representation capacity. Targeting at this issue, deep learning, especially the deep convolutional neural network (CNN) [15], provides a feasible way to improve the feature representation based on the end-to-end learning strategy, resulting in a breakthrough in many vision tasks. The existing studies reveal that the performance of deep models on remote sensing scene classification prefers a great depth, but the associated training difficulty must be alleviated by elaborating the model architectures, which require a large amount of practical experience [42], [43]. However, it is, to date, a mystery about how to design an effective architecture in a principled way. Roughly stacking the model layers would not consistently bring the performance improvement, but yields an increasing time cost. At this regard, neural architecture search (NAS) tries to search the optimal or near-optimal model architecture from a series of candidates in an automatic fashion instead of by manual design [44]. The goal is generally to find an effective network cell, which also has a small number of parameters to encourage high efficiency during model inference. Elsken *et al.* [46] have already been deeply investigated for the general image classification task and resulted in impressive performance. Considering the difference of the characteristics between natural images and remote sensing images [47], the existing methods [42] may produce suboptimal performance when directly applying on the remote sensing data. Hence, we are motivated to explore the effective model architectures for remote sensing scene classification through NAS, which is expected to yield a series of backbones for other remote sensing tasks, such as object detection and object relation analysis.

Most of the existing NAS methods are based on evolutionary searching and reinforcement learning, in which cases the search space of the model architecture is not differentiable. In general, thousands of architecture candidates are involved in the searching process, and each candidate is requested to be tested on the validation data, hence bringing considerable time cost and computational cost. To improve the searching efficiency, the differentiable architecture search (DARTS) [8] is proposed to speed up the searching process via a formal objective that could be solved by the gradient descent algorithm. While DARTS yields a pleasing performance on efficiency,

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this method has intrinsic issues that limit the searched results. In particular, by fixing the model depth during searching, DARTS could easily fall into a suboptimal solution on the cell architecture, which is coincident with the predefined depth. Moreover, as the architecture searching and the weight training processes continue alternatively, DARTS tends to remain the skip-connection and non-connection operations, where effective convolutional operations are dropped. This leads to a low capability of feature extraction by the resultant cells, which is addressed by, for example, Progressive Differentiable Architecture Search (P-DARTS) [9] that progressively increase the number of layers and channels and Partial Channel Connections Differentiable Architecture Search (PC-DARTS) [10] that stochastically sample parts of channels to be involved in searching while the rest channels remain unchanged.

High-quality and large-view remote sensing images contain diverse constructions and fields that exhibit rich structures, textures, and details [48]. The semantics of the scene are highly dependent on the objects and the background. But, the difficulty says that the scales of the objects could be extremely diverse, and the high-quality property results in different appearances of objects in the same class, as shown in Fig. 1. In addition, the large view also poses a problem that not all objects in the image are closely related to the scene semantics. These problems are particularly important and should be carefully considered in the design of CNN architectures, because the deep features of the CNN models may lose the information of the details (e.g., small objects) by improper pooling operation. Therefore, it is necessary to fuse the features with different-level semantics, which has already been integrated in the existing manually designed architectures [25] but not considered in an NAS process.

A natural way to accomplish the abovementioned target is to involve the multiscale fusion strategies in the searching space and find the best combination of different-level semantics, which is not a trivial task. The NAS methods, including DARTS and P-DARTS, build up the CNN architecture by cascading the searched cells and ignore the influence caused by different connections among different layers. This is mainly because the connection choices in a deep model are considerable, and the efficiency of a rough searching process is unacceptable. But, targeting at the remote sensing scene classification, the characteristics of the images facilitate us to employ a progressive bottom-up fusion strategy to search the connections, which deposits an acceptable cost on either computation or time.

As mentioned earlier, parts of the objects or the background in a remote sensing image may be uncorrelated with the scene semantics. This also causes that the scenes with similar appearances belong to different categories, while the scenes with different appearances have the same label. Regarding this specific issue, attention is a typical way to enhance the correlated features while suppressing the uncorrelated features in the deep models. In particular, the spatial attention helps the model to focus on the features spatially locating on the relevant object area, while the channel attention aims at selecting the important features corresponding to certain filtering patterns. Combining these two types of attention has already been verified to be effective in the manually designed models [20], but not in NAS yet. This motivates us to enrich the search space with the spatial attention and the channel attention, such that the searched model could be competent for high-performance remote sensing scene classification.

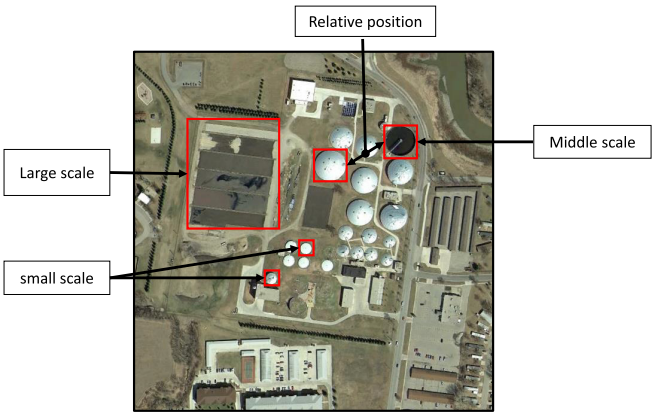


Fig. 1. Illustration of multiscale in remote sensing images.

Based on the abovementioned discussions, in this article, we propose a novel NAS method that involves the search of the attention operator and the multiscale fusion operator for deep learning-based remote sensing scene classification. The main contributions of this work are summarized as follows.

1) We develop a progressive searching method to build the bottom-up multiscale fusion architecture in the NAS process. The resultant model takes the advantages of low-level, mid-level, and high-level features in the classification, and this method can alleviate the issue of unstable architectures caused by undue iterations of searching.
2) We propose to enrich the search space with the attention operators, including the spatial attention, the channel attention, and their combination. This could help the searched models to identify the relevant objects in the remote sensing images and, hence, improve the classification performance.
3) The superiority of the proposed method is demonstrated through extensive ablation studies and comparisons with the existing manually designed deep models on the popular datasets.

The rest of this article is organized as follows. We review the related literature to this work in Section II. Section III introduces the proposed NAS method for remote sensing scene classification in detail. Experiments and discussions are presented in Section IV, with the conclusions drawn in Section V.

II. Related Work

In this section, we briefly review the literature related to this work.

A. Auto Deep Learning

Deep learning has received extensive attention in recent years. The manually designed deep models promote the performance of various vision tasks on natural images, which bring a difficult job on architecture design. It is not a trivial

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task to build up an effective neural network architecture manually, where lots of experiences on different architectures are required. Fortunately, the auto deep learning, or the NAS, provides a promising way to automatically find proper architectures that could yield pleasing performance, alleviating the insufficient experience problem. The existing NAS methods could be grouped into the following three classes.

1) Evolutionary-Based NAS: This type of methods tries to encode different operators of the neural network, which are used in the evolutionary algorithms to select the best architecture on a specific task. Liu *et al.* [12] were the first trial to employ the evolutionary method on NAS. Real *et al.* [13] involved the age attribute into the tournament selection evolutionary algorithm. Since the evolution of the architecture in the search process is random, it is difficult to determine when the best candidate is acquired, which results in very long time for searching. In addition, the evolution of the model is independent between the searching process and the validation process, and hence, it is prohibitive to adapt the search stage based on the validation stage and vice versa.

2) Reinforcement Learning-Based NAS: The typical reinforcement learning scheme regards an action as an agent of the model architecture, where the searching space corresponds to the action space, and the validation performance of the model is viewed as the reword. Different trials have been devoted to the design of the agent and the optimization strategies. Baker *et al.* [14] implemented the learning process based on Q-learning and the greedy searching method to determine the operators layer by layer. Gong *et al.* [15] employed the recurrent neural network as the agent and optimized the architecture using cascade searching. Pham *et al.* [16] developed a weight-shared supergraph as the basic network cell, which was optimized and then used to construct the whole architecture. The reinforcement learning-based methods have high efficiency on searching high-performance deep models. But, it is required to train the model from scratch in each searching iteration, which poses a great cost on hardware resources.

3) Gradient Optimization-Based NAS: The cells of the deep models can be viewed as a supergraph, which is constructed as a weighted combination of all candidate operations. By optimizing the weights based on certain criteria, the important operations become dominant to the task of interest, while the negligible operations are suppressed and then dropped, resulting in a compact yet effective network cell. The whole network is built up by cascading the searched cells. The earliest gradient-based idea was proposed in DART [8], which alleviated the issue of huge computational costs posted by the reinforcement learning algorithms and the evolutionary algorithms. Chen *et al.* [17] proposed to progressively increase the depth of searched architectures during the training procedure, which could suppress the difference of the performances between the search stage and the validation stage. Xu *et al.* [18] sampled the connections of the supergraph, generating a regularized subgraph that was optimized by the gradient descent strategy. This enabled the stability of the searching process. Peng *et al.* [42] considered to integrate four popular datasets to search the basic network cells and also employed a multistage progressive searching strategy to stabilize the searched cells. Remote Sensing Differentiable Architecture Search (RS-DARTS) [45] uses partial channel sampling in the search phase and adds noise to the skip connection operation weight. Effectively solve the problem of skip connection instability and reduce the cost of network search. These gradient optimization-based methods can, on the one hand, reduce the resource costs by formulating the connections as the optimizable graph weights, and on the other hand, the searching and validation processes are alternatively optimized, in which the output prefers task-oriented models.

B. Remote Sensing Scene Classification

The conventional remote sensing scene classification methods employ the hand-crafted visual features to represent the images, most of which care about the low-level features that have very limited capacity on modeling complex scene structures. To improve the feature representation, the low-level features are integrated as the mid-level features, such as the bag of visual word (BoVW) [19], the spatial pyramid matching (SPM) [20], the locality-constrained linear coding (LLC) [21], and the probabilistic latent semantic analysis (PLSA) [22]. These features extract abstract semantics from the low-level ones, which have improved discriminativity and corporate with the classifiers, such as support vector machine (SVM). The limitation of such a pipeline, i.e., integrating hand-crafted features with a classifier, is that the high-level semantics cannot be fully extracted, resulting in suboptimal classification performance. A direct way to address this issue benefits from deep learning, which provides an end-to-end learning schema for the feature extractors and the final discriminators. The existing deep models perform competitively compared with the conventional methods, not only in the field of computer vision, but also on the applications of remote sensing. Considering the difference between the characteristics of natural images and that of remote sensing images, the popular deep architectures are specifically adjusted to fit the remote sensing applications, as summarized as follows.

1) Deep Models Based on Transfer Learning: The natural images and the remote sensing images share similar structural and textual patterns, which provides the opportunity that the feature extractors for one type image can be fine-tuned to fit the other. This fine-tuning process is termed a simple form of transfer learning, which takes the advantages of the source domain that has a large amount of training data and alleviates the issue in the source domain that only contains a small amount of data. Hence, the transfer learning-based methods can be used in the remote sensing field where the collection of data is an expensive job. Han *et al.* [23] improved the performance of scene classification by fine-tuning the pretrained AlexNet model. Liu *et al.* [24] fine-tuned the hidden layers of the pretrained Visual Geometry Group Network (VGG) model based on high-resolution remote sensing images. Cheng *et al.* [2] employed the pretrained GoogleNet model that achieved pleasing performance.

2) Deep Models Based on Feature Fusion: The characteristics of remote sensing images state that the structures and

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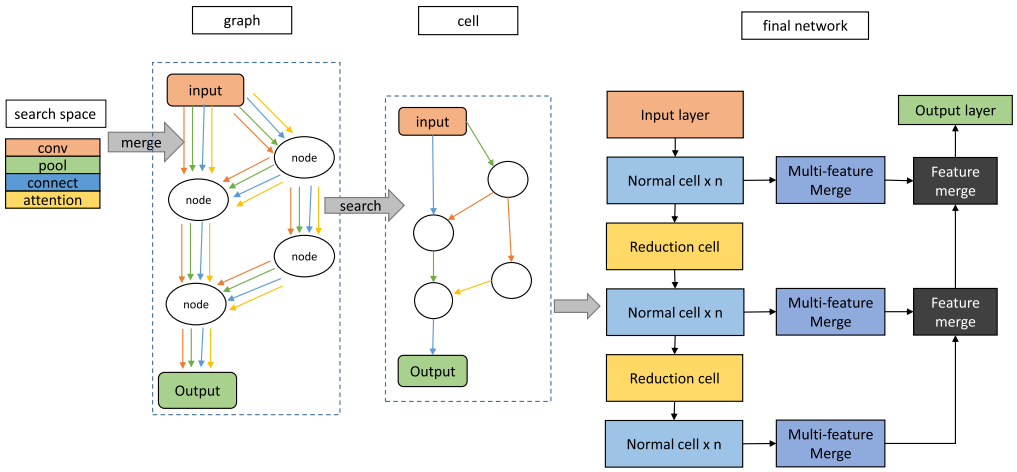


Fig. 2. Framework of the AF-NAS.

textures are complex, and the constructions and objects are with multiple scales. This imposes the necessity of high representation capacity for the feature extractors, which motivates the development of feature fusion. Chaib *et al.* [25] employed the canonical correlation analysis on the features of the VGG model for fusion. Liu *et al.* [26] proposed a two-stage method in which two transformation networks were used to fuse features. Lu *et al.* [27] developed a bidirectional adaptive fusion strategy to enhance the importance of low-level features. All these fusion methods could increase the utilization of multiscale features and improve the discriminativity of the features in foreground and background, hence achieving good performance.

3) Deep Models Based on Module Improvements: In remote sensing images, most of the objects exhibit the properties of small scales and indistinct appearances. The complex background could easily conceal the features of these objects. At this regard, the attention mechanism has been well investigated to increase the influence of the corresponding features. For example, Xu *et al.* [28] designed a novel network based on soft-attention and feedback attention. Wang *et al.* [29] employed recurrent attention to extract the high-level semantics and the enhanced spatial information. These attention-based methods can spatially locate the interested objects and, hence, are suitable for remote sensing data. Cheng *et al.* [30] tried to solve the problem of high intra-class variance and high inter-class similarity using metric learning, which also concerned the properties of remote sensing images. Guo *et al.* [31] took advantages of the generative adversarial networks that compensated the remote sensing images during training.

The abovementioned methods reveal that the elaborated architectures, including transfer learning, feature fusion, and module improvement, can yield pleasing performance on remote sensing scene classification, demonstrating that architecture is important. Therefore, in this work, we are well motivated to search the network architectures in a principled way, e.g., searching the attention operations and the fusion strategies in the NAS process.

III. Method

In this section, we introduce the proposed method in detail.

A. Framework

Considering the characteristics of remote sensing images, i.e., complex semantics and multiscale objects, we propose a novel NAS method for remote sensing scene classification, termed attention and fusion NAS (AF-NAS). This method aims at generating the best[1] architecture that can adaptively enhance the important features of the scene and selectively integrate the semantics of different scales, which is achieved by involving the candidates of attention and fusion mechanisms in the searching process. The whole framework is illustrated in Fig. 2. Following the existing works, the model is built up based on two types of cells, where one is the normal cell, and the other is the reduction cell. Both types of cells follow the same manner of graph construction, in which each node represents a feature map, and each edge denotes one of the candidate operations. To make the gradient optimization applicable, we assign each edge with a weight α, which is computed by softmax to ensure that the weight belongs to $[0, 1]$. In this way, the whole model is represented with the model weights ω (e.g., the parameters in convolutional layers) and the operation weights α. The searching process is then formulated as a second-order optimization problem, which can be simplified as two first-order optimization problems. That is, we can update α by fixing ω and then update ω by fixing α. The main contributions in the abovementioned NAS process consist of three core modules. In particular, the first module conveys that the searching space is equipped with the attention dimension. The second module targets at the characteristics of remote sensing images and is used to fuse the feature maps of different scales. The third module is a multistage progressive searching strategy, in which the top-down fusion process is divided into multiple stages where each stage only focuses on the corresponding layers of the whole network. The details of these three modules are presented in the following.

[1]The *best* indicates that the architecture yields the best performance according to the specified measurement in experiments.

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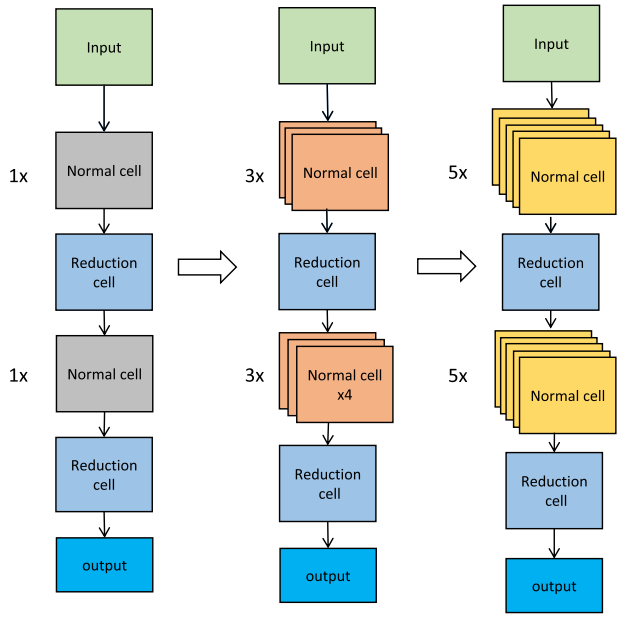


Fig. 3. Search strategy of P-DARTS.

TABLE I
SEARCHING SPACE

Operation	Candidate Shapes
Dilated Separable Convolution	3x3, 5x5
Separable Convolution	3x3, 5x5
Average Pooling	3x3
Max Pooling	3x3
Channel Attention	1x1
Spatial Attention	1x1
Cascading Spatial and Channel Attention	1x1
Parallel Spatial and Channel Attention	1x1
No operation	0
Skip-connection	1

In the network level search strategy after building the basic network unit, the method in this article is based on improvement of P-DARTS. As shown in Fig. 3, the P-DARTS allows the depth of searched architectures to grow gradually during the training procedure to improve the stability of searching and progressively delete some operations with low weights at the end of each stage to reduce the computation and memory usage.

B. Searching Space With Attention

The attention mechanism aims at enhancing the intermediate features extracted by the deep model. While this mechanism can be imposed on all layers, the involved computation is not negligible, and the resultant performance cannot be guaranteed. Hence, a remained issue is that which layer should be equipped with the attention operation. At this point, we propose to expand the searching space with the attention dimension, resulting in the full space as summarized in Table I. As seen, the channel attention, the spatial attention, and the cascading and parallel combinations are compensated as new dimensions in the space. We expect that the searching process can automatically determine where and which attention should

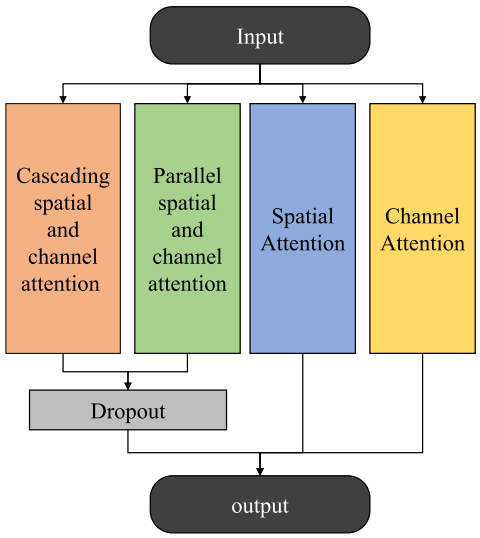


Fig. 4. Cell graph for attentions.

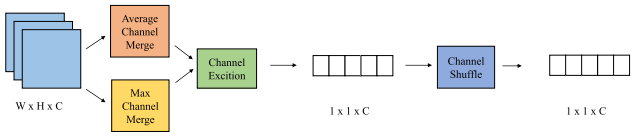


Fig. 5. Channel attention mechanism.

be used to improve the feature representation and the classification performance. The cell graph of the attentions is illustrated in Fig. 4.

We introduce four attention mechanisms in our search space, mainly channel attention mechanism [32], spatial attention mechanism [33], cascading spatial and channel attention mechanism [33], and parallel spatial and channel attention mechanism [34].

1) Channel Attention Mechanism: As shown in Fig. 5, the channel attention mechanism first performs global average pooling and global maximum pooling from feature dimensions, extracts the global semantics of each channel, and then performs nonlinear activation and channel-by-channel fusion on the extracted features, and finally passes the full. The connection layer reorganizes the feature map. The channel attention mechanism can mine the important information of each feature map in the channel dimension and enhance the network's attention to the semantic information of the global background of the image and the semantic information of the key objects in each scene.

2) Spatial Attention Mechanism: As shown in Fig. 6, the spatial attention mechanism first averages and maximizes the entire feature map from the channel dimension to obtain the average feature heat map and the maximum feature heat map, and then uses 1×1 convolution to pair the two feature heat maps of the two channels weighted average, and finally obtain the spatial attention feature map. The spatial attention mechanism can discover the important semantic information of each feature map in the spatial dimension and enhance the

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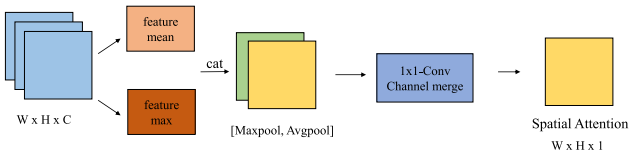


Fig. 6. Spatial attention mechanism.

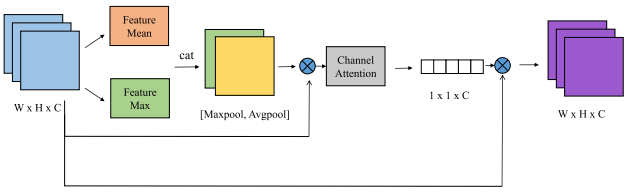


Fig. 7. Cascaded attention mechanism.

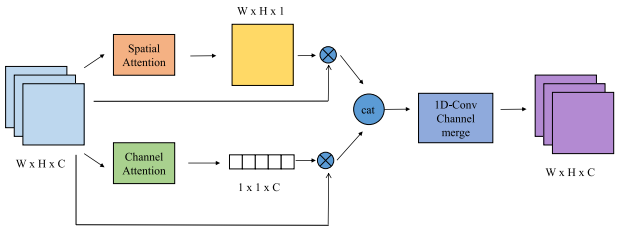


Fig. 8. Parallel attention mechanism.

network's attention to the local key background information of the image and the semantic information of the key small target and key object scenes.

3) Mixed Attention Mechanisms: Mixed attention mechanisms construct various mixed feature enhancement modules by cascading and paralleling the abovementioned two attention mechanisms. Mixed attention mechanisms combine two complementary attention mechanisms to construct a 3-D attention map by combining the local attention of the spatial angle and the global attention of the channel. As shown in Fig. 7, it is the cascaded attention mechanism. The cascaded attention mechanism first uses the spatial attention mechanism on the output feature map of the previous stage to obtain the spatial attention heat map and then performs the Hadamard product of the heat map with the original feature map. The spatially enhanced feature map is obtained, and then, the spatially enhanced feature map is input to the channel attention mechanism to obtain the entire 3-D complementary channel spatial attention heat map, and finally, the heat map and the input feature map are global product to obtain the attention-enhanced feature map.

4) Parallel Attention Mechanisms: As shown in Fig. 8, the parallel attention mechanism first inputs the output feature map of the previous stage into the spatial attention mechanism and the channel attention mechanism to obtain the spatial attention heat map and the channel attention heat map and then outputs the feature map in the same way as the previous stage. After the Hadamard product and the global product are performed, two enhanced feature maps will eventually be obtained. In this article, the two enhanced feature maps are filtered by convolution with channel weighting to obtain a 3-D global feature map, and finally, the enhanced output attention enhancement feature map is obtained.

The four attention mechanisms in Table I have different representational capacities. We empirically find that the searching process prefers the cascading spatial and channel attention, and the resultant architecture is usually equipped with a simple stack of this attention, whereas the other attentions are rarely selected. This is, perhaps, because the cascading spatial and channel attention has deeper layers and higher complexity than the other attentions. However, this fact is not expected because of the limited diversity of the selected operations, and the searched model is easy to overfit on the training data. To alleviate this issue, we propose to impose the dropout after the combined attentions and $L2$ regularization in the loss function. In addition, it is known that the skip-connection operation could be easily selected during the conventional searching process, but we find that the spatial attention and the channel attention can suppress the selection of this operation. Hence, we do not constrain these two types of attentions. Formally, the loss function of NAS in our method can be written as

$$L = \frac{1}{n}\sum_{i}\sum_{c=1}^{n}(\log(f(x_{ic}))y_{ic}) + \lambda_n\|\alpha_{\text{search}}\|_2^2 + \lambda_a\|\alpha_{\text{att}}\|_2^2 \tag{1}$$

where n is the number of training samples, $f(\cdot)$ is a candidate deep architecture, x_{ic} and y_{ic} denote the representation of each sample, α_{search} is the learnable parameters of the deep architecture, α_{att} is the weights corresponding to the combined attentions, and λ_n and λ_α are the regularization parameters.

C. Layer Connection for Feature Fusion

The features extracted in a single layer of the deep CNN model have insufficient ability on discriminating the objects of different scales, which motivates the feature fusion strategy in remote sensing scene classification. In particular, it is essential to exploit the features of multiple layers with different level semantics, such that the final classification layer can witness discriminative features for accurately predicting the label of the complex scene. Hence, in the proposed method, we employ a multistage and multiscale fusion manner to extract as much semantics as possible from the feature extractor backbone.

The fusion manner follows a top-down pyramid architecture with multiple stages, as shown in Fig. 9. The stages are defined between the reduction cells, i.e., all normal cells with the same spatial dimension belonging to a stage. In a certain stage, we concatenate the output feature maps of all normal cells, followed by 1×1 convolution to reduce the channel dimension. The reduction factor is 0.5, denoting that the output channel number is 0.5 times of the input channel number. Following a common model setting, the architecture has three stages, including the low-level stage, the mid-level stage, and the high-level stage. To ensure the presentation of the features of small objects, we implement the top-down fusion by fusing the lower-level stage with the higher-level stage. Upsampling

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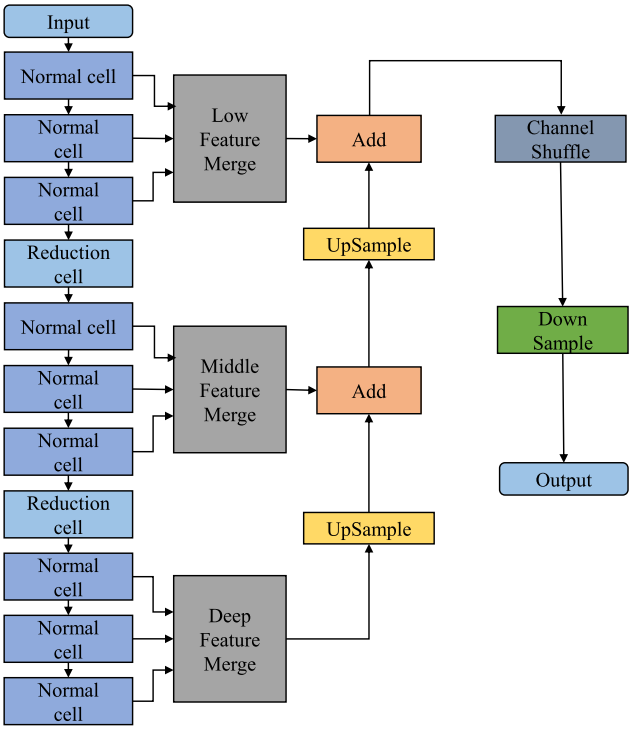


Fig. 9. Top-down feature fusion strategy.

is used to make sure that the feature map sizes of the two stages are compatible with each other. The fused feature maps are then processed by channel shuffle and downsampling, generating the final features for classification.

D. Progressive Searching

In the gradient-based NAS methods, the cell architectures are searched, while the whole model is built up by repeating the determined cells in each stage. We know that the information flow in the model may affect the node operations and the connections. Hence, when the model is equipped with the fusion architecture, the graph in the cells may have varied structures. To stabilize the searching process and obtain effective cells, we develop a progressive searching strategy, which is consistent with the top-down fusion manner, as shown in Fig. 10. In particular, we first search the cell structures in the high-level stage by only equipping the model with the high-level fusion part, where the mid-level and low-level fusion parts are dropped. Then, the cell structures in the mid-level stage are optimized by remaining the mid-level and high-level fusion parts. At last, the cell structures in the low-level stage are searched by arming with the whole feature fusion structure, resulting in the final deep architecture.

IV. Experiments

In this section, we present the experimental studies to validate the effectiveness of the proposed AF-NAS method.

A. Experimental Settings

1) Datasets: To thoroughly examine the performance and generalization ability of the proposed method, we select four

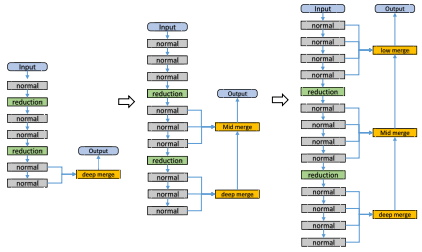


Fig. 10. Progressive searching strategy.

popular remote sensing image datasets, which are listed as follows.

1) *Aerial Image Dataset (AID) [35]:* This dataset contains 10 000 remote sensing scene images of 30 classes that are collected from the Google Earth platform. The sensing resolution ranges from 1/2 to 8 m. The image resolution is fixed to 600 × 600. Each class consists of 220–420 images.
2) *Northwestern Polytechnical University (NWPU) Dataset [2]:* This dataset collects 31 500 images of 45 classes with each containing 700 images, which are also sampled from the Google Earth data. The sensing resolution ranges from 0.2 to 30 m, while the sizes of the images are 256 × 256.
3) *Remote Sensing Image Classification Benchmark (RSI-CB256) Dataset [36]:* The samples in this dataset are obtained from multisource sensing devices. The sensing resolution is from 0.3 to 3 m. There are totally 24 000 images of 35 categories, with the image size of 256 × 256.
4) *PatternNet Dataset [37]:* This dataset is collected using multiple different sensing devices, containing 30 400 high-quality images of 38 categories with each having 800 samples. The sensing resolution ranges from 0.062 to 4.693 m, and the image size is fixed to 256 × 256.

These datasets exhibit various sensing conditions and scene contents, imposing a challenge for robust remote sensing scene classification.

2) Evaluation Metrics: To quantitively evaluate the performance, we employ three metrics for comparison.

a) Overall accuracy: The overall accuracy (OA) is computed as the ratio of the correct predictions with respect to all predictions, which is expressed as

$$OA(\%) = \frac{N_C}{N_A}. \tag{2}$$

b) Confusion matrix: The confusion matrix (CM) is a matrix with the horizontal dimension indicating the predicted labels and the vertical dimension indicating the ground-truth labels. The percentages in the matrix denote how many samples of a certain class (in the vertical dimension) are predicted as each of all the classes (in the horizontal dimension). This metric not only depicts the accuracy on each class, but also illustrates the discrimination of which two classes are easily collapsed.

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c) Average accuracy: The average accuracy (AA) represents the average proportion of the classification accuracy of each category sample to the overall category, which is expressed as

$$AA(\%) = \frac{1}{N}\sum_{i=1}^{n}\alpha_i. \tag{3}$$

3) Searching Settings: Recall that the proposed architecture has three stages, including the low-level, mid-level, and high-level stages. In these stages, we set the numbers of cells as 5, 9, and 12, respectively, and set the initial channel numbers as 8, 14, and 20, respectively. As shown in Fig. 2, each cell contains $N = 11$ nodes, including two input nodes, eight intermediate nodes, and one output node. According to the edge selection rules in P-DARTS [38], the clipping numbers of the operations in the intermediate nodes are set to 1, 2, and 3 in the stages, respectively. Since the searching process is stage by stage, the iterations used for searching the three stages are 15, 25, and 30, respectively. The ratios of the utilized data in both training and validation are set as 90%, in which way we could acquire more and more accurate orders of the α parameters along the searching loops.

In the searching phase, the input sizes of the training images are fixed to 64×64. The general data augmentation methods are employed to enrich the training data and avoid overfitting, such as random cropping, flipping, and rotation. We implement the proposed methods under the Pytorch framework on a device with two NVIDIA RTX2080Ti GPUs. The Adam algorithm is used to optimize α, with the settings that the batch size is 32, the learning rate is 0.0005, the weight decay is 0.001, and the momentum is (0.5, 0.999). Stochastic gradient descent (SGD) is employed to update the model weights, where the initial learning rate is 0.025, the momentum is 0.9, and the weight decay is 3×10^{-4}.

In the evaluation phase, the input sizes of the training images are fixed to 64×64. The number of operations in each node is set to 2. The searched cells are embedded in the designed architecture in Fig. 9, generating the final model. In particular, the model has the initial channel number of 64, 10 normal cells, and 3 reduction cells. To evaluate the model on each dataset, the training process starts from scratch and ends in 500 epochs. The batch size is set to 64, and the other settings follow those in the searching phase. In addition, we also employ the gradient clipping technique with the clipping threshold of 16 and the dropout layer with the probability of 0.2.

B. Searching Space

We first discuss the searching space of the proposed method, which has more candidates than the original DARTS method. Table I lists the involved 12 operations. As mentioned earlier, each cell contains seven nodes, with two input nodes accepting the data from the previous cells and one output node processing the features extracted from the intermediate nodes. The number of edges to be searched in a cell is $2+3+4+5 = 14$. Taking all operations into account, it will result in $14^{12} \approx 10^{14}$ candidate graphs in each cell. The two types of cells bring $(10^{14})^2 = 10^{28}$ candidates. In practice, to suppress the searching space, we only keep two edges that correspond to the two highest weights, and the cell has $\prod_{l=1}^{4}(((l \times (l+1))/2) \times 12^2) \approx 10^{10}$ candidate graphs in the searching phase. Considering that we search two types of cells, i.e., normal cell and reduction cell, the number of all candidates increases to 10^{20}. The proposed method allows us to reduce the number of candidates in each searching stage, hence largely decreasing the computational cost. As shown in Table II, the searching process with a single GPU takes 42 h (1.6 GPU days) in the AID dataset and 44 h (1.8 GPU days) in the NWPU dataset.

TABLE II
SEARCH COSTS AND PARAMETERS OF DIFFERENT METHODS ON AID DATASET

Method	Search Cost (GPU days)	Parameter(M)
ENAS	3.0	4.8
DARTS	5.0	3.4
P-DARTS	2.0	3.5
PC-DARTS	2.6	3.0
Our method	2.8	3.8

C. Searched Models

Here, we show the searched architectures for the normal cell and the reduction cell on the selected datasets, as shown in Fig. 11. The stride of convolution in the normal cells is set to 1, and that in the reduction cells is set to 2 to halve the size of the output feature maps. As seen from the architectures, the proposed method tends to select deep connections (such as attentions), which is different from DARTS that prefers shallow connections (such as skip connection). This drives our method to preserve deep layers in the searched architecture, which, hence, has stronger representational capacity compared with the model searched by DARTS.

D. Comparison With the State of the Arts

To examine how the searched architectures perform on the public datasets, we make comparisons with the manually designed deep models, which have pleasing performance on visual classification tasks. In particular, three deep CNN models are selected, including VGG-16 [38], GoogleLeNet [39], and ResNet-50 [40]. For these competitors, the ratios of training data on the AID, NWPU, RSI-CB, and PatternNet datasets are set as 50%, 60%, 60%, and 50%. For our proposed method, we set up two training cases with the training ratios as 40% and 60%, respectively. The comparison results are presented in Table III, where OA is used as the evaluation metric. It is observed that the proposed method achieves notable improvement over the other popular deep models, even including the fine-tuned models.

Next, we illustrate the confusion matrices on each dataset. Fig. 12 shows the CM of AID when the training-test split is 40%:60%. As seen, the accuracies on most classes are above 90%. The accuracies on Center, Church, Park, and Resort are above 85%. The worst case occurs on the School class that has the accuracy of 77%, and 6% of the images belonging to

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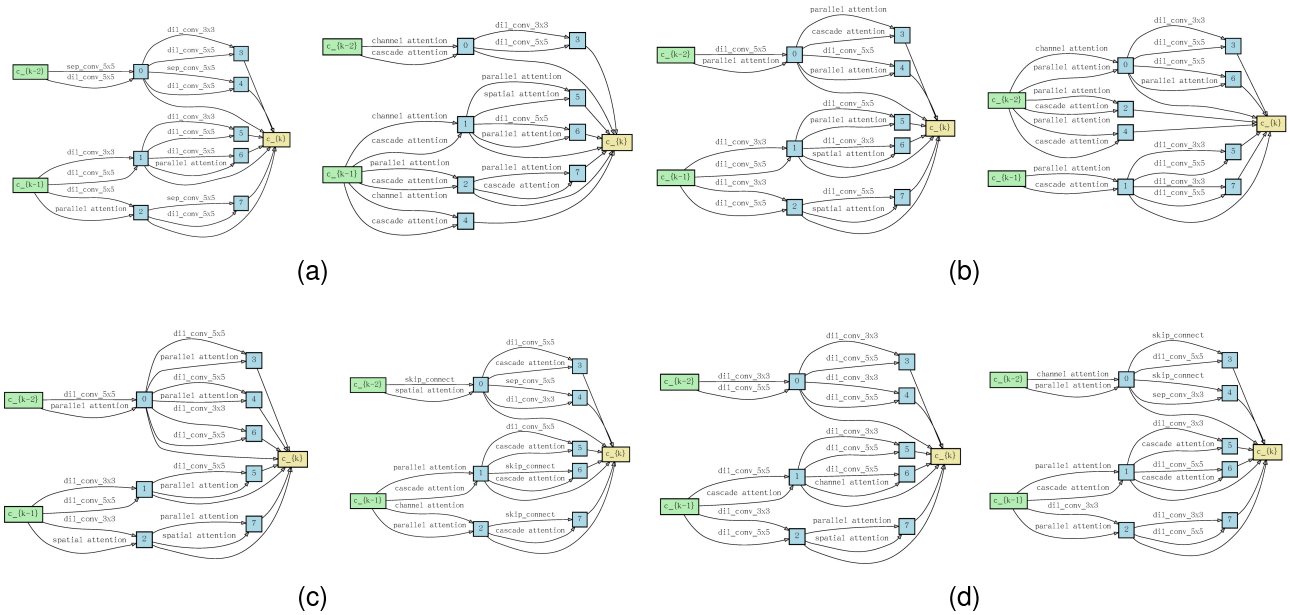


Fig. 11. Searched architectures of cells on different datasets. (a)–(d) Normal cells and reduction cells on AID, NWPU, RSI-CB, and PatternNet datasets, respectively.

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE PUBLIC DATASETS

Deep Models	AID (%)		NWPU (%)		RSI-CB (%)		PatternNet (%)	
	OA	AA	OA	AA	OA	AA	OA	AA
VGG-16	88.56	87.98	85.97	85.29	96.53	95.23	97.31	96.23
GoogLeNet	84.27	84.23	83.29	83.03	96.21	96.21	96.12	95.81
ResNet-50	86.94	86.21	86.96	85.98	97.09	97.05	96.71	95.97
VGG-16(fine-tunning)	92.03	91.58	91.32	89.93	97.26	97.02	98.31	98.23
GoogLeNet(fine-tunning)	90.31	89.69	89.42	88.20	98.14	98.10	97.56	97.52
ResNet-50(fine-tunning)	92.14	91.98	91.63	90.92	98.12	97.52	98.23	97.95
Our Searched model(40%)	93.96	92.31	93.46	91.36	99.13	98.51	99.47	99.44
Our Searched model(60%)	95.65	93.47	95.32	93.73	99.30	98.86	99.51	99.52

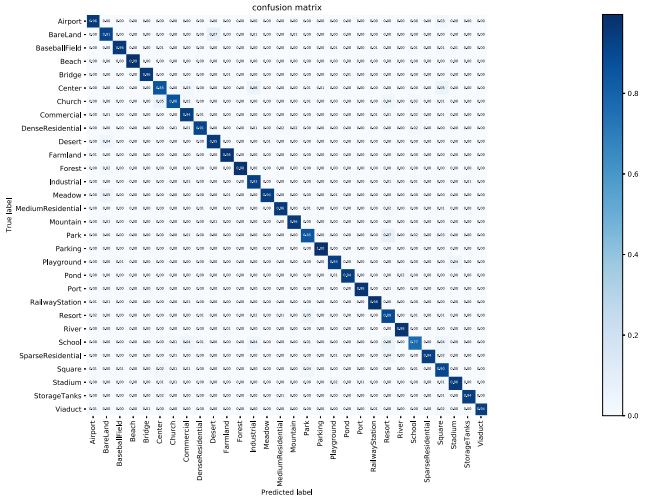


Fig. 12. CM on AID.

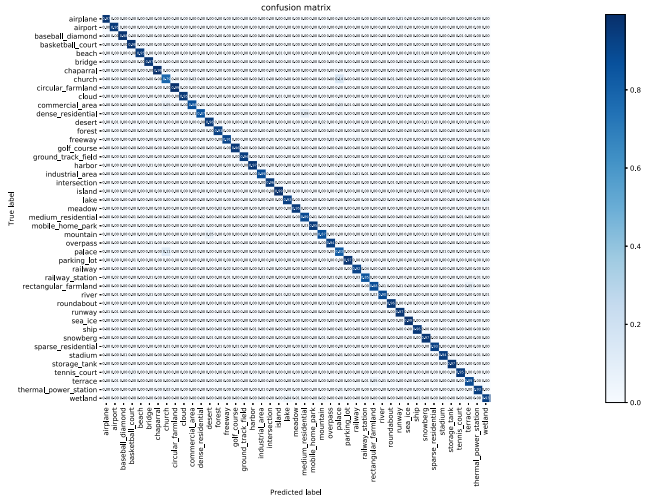


Fig. 13. CM on NWPU.

School class are mistakenly classified as Resort, which reveals that School and Resort have high inter-class similarity.

On the NWPU dataset, the CM obtained when the training-test split is 4:6 is shown in Fig. 13. It is seen that most categories exhibit the accuracies over 88%, while the other categories lie in [83%, 88%]. The Church class is easily misclassified as the Palace class.

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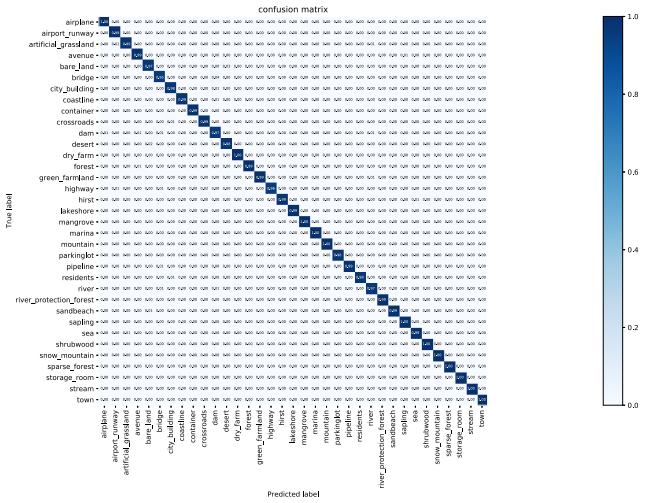


Fig. 14. CM on RSI-CB.

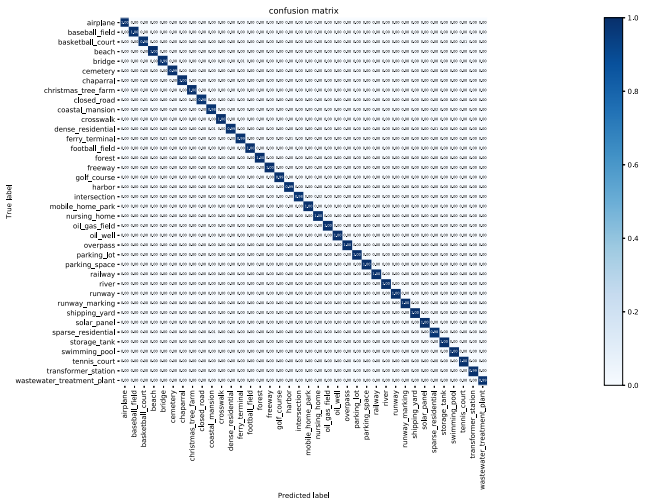


Fig. 15. CM on PatternNet.

Fig. 14 shows the CM on RSI-CB when the training-split ratio is 4:6. The proposed method has very pleasing performance on this dataset, where the accuracies on almost all classes are above 97%. The bridge class and the dam class are difficult to be discriminated.

Fig. 15 presents the CM on PatternNet while the training ratio is 40%. We also see that our method performs well, and the results of all classes are over 98%.

We also make a comparison between the proposed method and the existing NAS methods. The training-test split is 6:4 for all datasets and all competitors. The results are listed in Table IV, which sufficiently verifies the effectiveness of our method. The searched model has higher ability of feature extraction and feature fusion, yielding improved performance for remote sensing scene classification.

Our method achieves better results than DenseNet, Efficient Neural Architecture Search (ENAS), and DARTS, which proves that the fusion strategy and attention operation are advanced in remote sensing scene data analysis. Moreover, compared with P-DARTS and PC-DARTS, our method has a better performance, which shows the superiority of our network search strategy in remote sensing image scene classification.

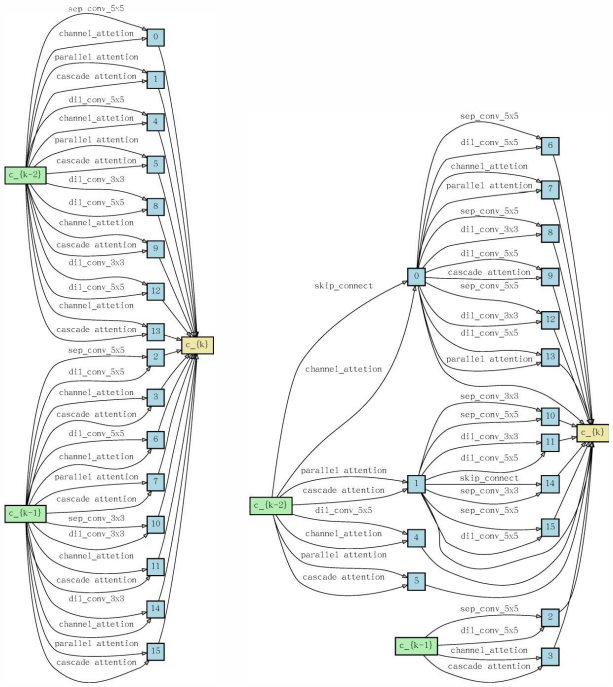


Fig. 16. Normal cell and the reduction cell searched by P-DARTS.

E. Ablation Studies

The performance of the proposed method is correlated with the feature fusion strategy, the number of cells, the searching strategy, and the attention structure. Here, we conduct ablation studies to examine how these factors affect the whole architecture. All the following experiments are conducted on the AID dataset, while the training-test split is set to 6:4.

1) Investigation on Feature Fusion: We first examine the performance when the model is equipped with and without the multistage and multiscale feature fusion strategy. The results are shown in Table V, which indicates that the proposed fusion strategy is useful for recognizing the remote sensing data.

2) Investigation on the Number of Normal Cells: We examine how the number of the normal cells affects the performance of the searched architecture. In particular, the tested number ranges from 8 to 13, producing the results as shown in Table VI. As observed, the best performance is obtained when the number is 10.

3) Investigation on the Progressive Searching Strategy: The progressive searching strategy is the key to improve the stability and effectiveness of the architecture. We, thus, validate the proposed searching strategy by comparing with the conventional methods. The results of the proposed method, P-DARTS, and DARTS are illustrated in Figs. 10(a), 16, and 17, respectively.

It is seen from Fig. 16 that P-DARTS tends to select multiple operations to expand the width of the cell, whereas duplicated operations may exist in the resultant graph. Since

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TABLE IV
COMPARISON BETWEEN THE PROPOSED METHOD AND THE EXISTING NAS METHODS

Network	AID		NWPU		RSI-CB		PatternNet	
	OA	AA	OA	AA	OA	AA	OA	AA
DenseNet-169 [41]	91.86	92.06	90.36	90.21	97.56	98.51	97.61	97.41
ENAS [16]	93.89	94.03	93.54	93.21	97.65	97.32	98.03	98.91
DARTS (single-stage)	94.24	94.21	93.04	93.19	97.92	98.21	98.12	98.25
P-DARTS (multi-stage)	94.84	94.62	94.09	94.81	98.23	98.50	98.33	98.23
PC-DARTS (sample super-graph)	94.63	94.13	94.34	95.16	99.13	99.34	98.34	99.02
RS-DARTS(partial channels)	94.14	94.12	93.68	94.48	99.42	99.44	99.52	99.32
Our method	95.65	93.47	95.32	93.73	99.30	98.86	99.51	99.52

TABLE V
INVESTIGATION ON THE FEATURE FUSION STRATEGY

Settings	OA (%)
With feature fusion	95.65
Without feature fusion	94.64

TABLE VI
PERFORMANCE OF THE SEARCHED ARCHITECTURE WITH DIFFERENT NUMBERS OF NORMAL CELLS

#normal cells	OA (%)
8	94.82
9	95.03
10	95.65
11	95.21
12	94.96
13	94.25

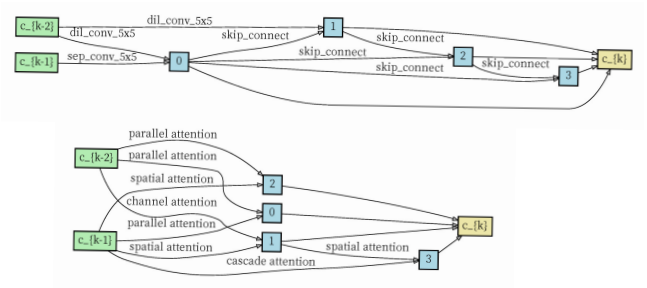


Fig. 17. Normal cell and the reduction cell searched by DARTS.

the depth of the cell is not sufficient, the final performance is, thus, limited. In the case of DARTS, the training process is early stopped to avoid too many skip connections in the resultant architecture. We observe from Fig. 17 that DARTS still prefers to select the skip connections instead of the attentions in the normal cell. The reduction cell only involves a series of attentions without any convolutions. This reveal that DARTS is not stable to perform an effective search in the attention-enabled searching space, since too many skip connections have a bad effect on nonlinear representation ability of the model. The quantitative comparison can be found in Table VII.

4) Investigation on the Dropout Layer in Attention Structure: Considering that the designed attention structure has a dropout layer upon the cascading attention and the parallel attention, we examine the performance of the structure with and without the dropout layer. The searched cells without and

TABLE VII
PERFORMANCE COMPARISON OF DIFFERENT SEARCHING STRATEGIES

Search Strategy	OA (%)
Progressive Searching	95.65
P-DARTS	94.92
DARTS with early stop	87.5

TABLE VIII
PERFORMANCES OF THE SEARCHED ARCHITECTURES WITH AND WITHOUT THE DROPOUT LAYER

Options	OA (%)
With Dropout	95.65
Without Dropout	93.13

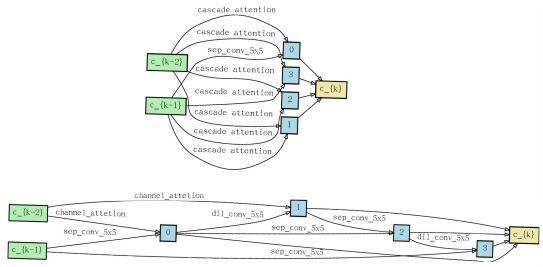


Fig. 18. Normal cell and the reduction cell searched without the dropout layer.

with dropout are illustrated in Figs. 10(a) and 18, respectively. The comparison indicates that the normal cell searched without dropout employs a parallel of multiple cascading attention, while the reduction cell searched without dropout rarely uses attentions. This is not expected, since the representation capacity of such architectures is limited, which can be observed from the quantitative comparison in Table VIII. By contrast, the option with the dropout layer yields higher performance.

V. CONCLUSION

NAS plays an important role on designing effective deep models for various vision tasks. But, the existing NAS methods do not fully consider the characteristics of remote sensing data. At this point, we explicitly model the attention-enabled search space and the feature fusion strategy in the NAS process to address the issues caused by multiscale objects and complex backgrounds in remote sensing images. Although the difficulty of architecture search increases with the increase in search space complexity, we add $L2$ regularization operation to punish mixed attention operations in the loss function.

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The searched architecture can extract features with improved representational capacity, thus yielding pleasing performance on scene classification. To stabilize the searching process, we also propose a progressive searching strategy to search the cells stage by stage. The experiments on four popular remote sensing datasets validate the effectiveness of the proposed method.

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